Homework 3

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Exercise 3.1

Let $n \geq 2$ be a positive integer, and write $M_n(\mathbb{C})$ for the associative algebra consisting of all $n \times n$ complex matrices. Prove that $M_n(\mathbb{C})$ is simple.

Proof. If $\mathfrak{h}$ is a nonzero ideal of $M_n(\mathbb{C})$, we shall prove that $\mathfrak{h} = M_n(\mathbb{C})$. Take a nonzero element $A \in \mathfrak{h}$. Denote A by (a_{ij}). Since $\mathfrak{h}$ is an ideal, $a_{jk}E_{il} = E_{ij}AE_{kl} \in \mathfrak{h}$. Since $A \neq 0$, there exists j, k such that $a_{jk} \neq 0$. Then it follows that for each E_{il},

$$E_{il} = \left(a_{jk}^{-1}\right)\left(a_{jk}E_{il}\right) \in \mathfrak{h}$$

Finally, since $\mathfrak{h}$ is a subalgebra of $M_n(\mathbb{C})$ contaning all E_{il}, we know that

$$\mathfrak{h} \supseteq \operatorname{span}(E_{il})_{1 \leq i,l \leq n} \implies \mathfrak{h} \supseteq M_n(\mathbb{C})$$

The above implies that $M_n(\mathbb{C})$ is simple. □

Exercise 3.2

Let $n \geq 2$ be a positive integer, and write $\mathfrak{sl}_n(\mathbb{C})$ for the Lie algebra consisting of all trace zero $n \times n$ complex matrices. Prove that $\mathfrak{sl}_n(\mathbb{C})$ is simple.

Proof. If $\mathfrak{h}$ is a nonzero ideal of $\mathfrak{sl}_n(\mathbb{C})$, we shall prove that $\mathfrak{h} = \mathfrak{sl}_n(\mathbb{C})$. Take a nonzero element $A \in \mathfrak{h}$.

- If $A = \operatorname{diag}(a_1, \cdots, a_n)$ is diagonal, then there exists k, l such that $a_k \neq a_l$. Then

  $$[A, E_{kl}] = AE_{kl} - E_{kl}A = a_kE_{kl} - a_lE_{kl} = (a_k - a_l)E_{kl} \in \mathfrak{h}$$

  Thus

  $$E_{kl} = (a_k - a_l)^{-1}((a_k - a_l)E_{kl}) \in \mathfrak{h}$$

- If A is not diagonal. Let H be a nonzero diagonal element of $\mathfrak{sl}_n(\mathbb{C})$. Consider the operator $\mathrm{ad}(H)$

$$\begin{aligned} \mathrm{ad}(H) : \mathfrak{sl}_n(\mathbb{C}) &\to \mathfrak{sl}_n(\mathbb{C}) \\ Y &\mapsto [H, Y] \end{aligned}$$

Since $\mathfrak{h}$ is an ideal, for each $X \in \mathfrak{h}$, we have

$$\mathrm{ad}(H)(X) = [H, X] \in \mathfrak{h}$$

which implies that $\mathfrak{h}$ is an invariant space of $\mathrm{ad}(H)$. Then there exists an eganvector Y of $\mathrm{ad}(H)$, such that $\mathrm{ad}(H)(Y) = \lambda Y$. Suppose that $H = \sum_{k=1}^{n} h_k E_{kk}, Y = \sum_{i,j} y_{ij} E_{ij}$

$$\begin{aligned} \sum_{k,j} \lambda y_{kj} E_{kj} = \mathrm{ad}(H)(Y) &= \sum_{k=1}^{n} \sum_{j=1}^{n} y_{kj} h_k E_{kj} - \sum_{k=1}^{n} \sum_{i=1}^{n} y_{ik} h_k E_{ik} \\ &= \sum_{k=1}^{n} \sum_{j=1}^{n} y_{kj} \left(h_k - h_j\right) E_{kj} \end{aligned}$$

Then for each k, j, there must be

$$h_k - h_j = \lambda, \quad \text{or} , \quad y_{kj} = 0$$

Once we take $H = \sum_{i=1}^{n} h_i E_{ii}$, such that $h_i \neq h_j, \forall i \neq j$.

1. If $\lambda = 0$, since $h_k \neq h_j$ for all $k \neq j$, there must be $y_{kj} = 0$. Y is diaganol this situation.
2. If $\lambda \neq 0$, there at most one pair (k, j) such that $h_k - h_j = \lambda$, then $Y = y_{kj} E_{kj}$.

$$E_{kj} = y_{kj}^{-1} Y \in \mathfrak{h}$$

There always exists some $E_{kl} \in \mathfrak{h}$, where $k \neq l$. Since

$$[E_{ab}, E_{kl}] = \delta_{bk} E_{al} - \delta_{la} E_{kb}$$

Then for each $a \neq l$, we have

$$E_{al} = [E_{ak}, E_{kl}] \in \mathfrak{h}$$

And for each $b \neq k$, we have

$$E_{bk} = -[E_{lb}, E_{kl}] \in \mathfrak{h}$$

We can repeat this process finite times to generate all E_{mn} satisfying $m \neq n$.
Finally, we have for each $E_{kk} - E_{ll}$, $k = l$,

$$E_{kk} - E_{ll} = [E_{kl}, E_{lk}] \in \mathfrak{h}$$

Thus

$$\mathfrak{h} = \operatorname{span}\{E_{mn}, E_{kk} - E_{ll}\}_{m \neq n, k \neq l} = \mathfrak{sl}_n(\mathbb{C})$$

$\square$